
Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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Abstract

1 Every major transition in theoretical physics follows the same pattern: a known
2 classical law is extended by a correction term that vanishes at the classical limit.
3 Special relativity introduced the Lorentz factor as a multiplicative correction to
4 proper time intervals; Yukawa screening added an exponential decay to Coulomb’s
5 law; the isothermal expansion of an ideal gas introduces a logarithmic entropy cor-
6 rection. We ask whether this structural regularity can be exploited computationally
7 to recover corrections from noisy data. We present **ADCD** (Asymptotic Dictio-
8 nary Correction Discovery), a deterministic framework that searches a bounded
9 dictionary of five algebraically regularized primitive functions, each analytically
10 constrained to vanish at the classical asymptote, over a finite grammar. Algebraic
11 regularization eliminates catastrophic cancellation at classical limits, a persistent
12 numerical failure mode in standard regression at physical boundaries. Candi-
13 dates pass a cascade of physical gates (AST complexity, dimensional consistency,
14 transcendental safety, and asymptotic regime verification) before any numerical
15 fitting. A four-step validation protocol determines when data are sufficient to
16 support a structural claim. We validate on three canonical synthetic scenarios:
17 special-relativistic time dilation, Yukawa plasma screening, and ideal-gas isother-
18 mal expansion. Given human-specified dimensionless arguments, ADCD performs
19 confirmatory structural fitting and recovers the correct algebraic structure with
20 normalized mean-squared errors below 3.4×10^{-3} and BIC differences of 59.6–
21 2024 over the best competing alternative. All results are byte-exact reproducible
22 across independent runs. We report identifiability boundaries explicitly and make
23 no claims beyond the validated regime.

24 **Keywords:** symbolic regression; physical laws; correction discovery; dimensional
25 analysis; identifiability; Bayesian model selection

26 1 Introduction

27 Every major transition in theoretical physics follows a pattern: a known classical law is extended by a
28 correction term that vanishes at the classical limit. For example, Einstein’s special relativity Einstein
29 [1905] extended Newton’s invariant time by a multiplicative Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$,
30 which reduces exactly to 1 when $v \ll c$. Similarly, Coulomb’s force is screened by an exponential
31 factor in plasmas Debye and Hückel [1923], and ideal-gas isothermal expansion receives a logarithmic
32 entropy correction Callen [1985]. In each case, the correction is a structured function with a strict
33 classical limit.

Modern symbolic regression (SR) engines (e.g., PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020]) are powerful, but their unconstrained generality creates specific difficulties for correction discovery. First, they suffer from catastrophic cancellation at classical limits: trying to optimize $1 + \Delta$ as $\Delta \rightarrow 0$ leads to gradient collapse, even in `float64`. Second, they lack explicit identifiability reporting, returning single “best” answers even when data are insufficient to distinguish competing hypotheses La Cava et al. [2021]. Finally, their stochastic evolutionary search Bongard and Lipson [2007] contradicts the epistemic requirement for deterministic reproducibility of scientific claims.

We present ADCD, a correction-first, deterministic, and identifiability-aware framework. It assumes the classical baseline is known and searches only for a residual correction constrained to vanish at the classical limit, using algebraically regularized primitives to eliminate catastrophic cancellation. Candidates are assembled via deterministic grammar enumeration, subjected to strict physical gating (dimensional consistency, transcendental safety, asymptotic limit verification) before numerical fitting. ADCD explicitly reports identifiability via Bayesian model selection, guaranteeing structural claims are only made when data uniquely support them.

2 Related Work

General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce strict classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. Physics-informed neural networks (PINNs) Raissi et al. [2019], Karniadakis et al. [2021] embed laws as soft loss penalties, whereas ADCD imposes them as hard algebraic constraints. While dimensional analysis has been used in SR Tenachi et al. [2023], Petersen et al. [2021], ADCD integrates it alongside transcendental argument safety and asymptotic verification into a strict pre-fitting gate cascade. Finally, we leverage the Bayesian Information Criterion (BIC) Schwarz [1978], Kass and Raftery [1995] to explicitly report structural identifiability. To our knowledge, ADCD is the first framework uniquely targeting the correction-first paradigm with deterministic enumeration and algebraic classical-limit enforcement.

3 Methods

3.1 Problem formulation

Let y_{obs} be the observed target and y_{cl} the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & \text{(multiplicative)} \\ y_{\text{obs}} - y_{\text{cl}} & \text{(additive)} \end{cases} \quad (1)$$

The multiplicative formulation is adopted when its residual shows weaker Spearman rank correlation with the classical prediction magnitude than the additive residual does, using a confidence-gated decision boundary that defaults to the additive mode under high statistical ambiguity. ADCD searches for $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of physical inputs, $\boldsymbol{\theta}$ are continuous free parameters, and $\hat{\Delta} \rightarrow 0$ as the relevant dimensionless parameter $u \rightarrow 0$.

3.2 Regularized primitive dictionary

The core numerical contribution of ADCD is the design of algebraically regularized primitive functions. Each primitive $D(u)$ satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity, not a numerical approximation. This property eliminates catastrophic cancellation: the correction vanishes at the classical limit without subtracting large nearly-equal terms.

Table 1 lists the five primitives. The practical consequence is demonstrated for `D_1or`: in `float32`, the naive form $(1 - u)^{-1/2} - 1$ collapses to exactly 0.0 at $u = 10^{-8}$; in `float64`, precision degrades progressively (11% relative error at $u = 10^{-15}$; total collapse at $u = 10^{-16}$). The regularized form $u / [\sqrt{1 - u}(\sqrt{1 - u} + 1)]$ preserves full precision at all scales, returning 5.0×10^{-17} at $u = 10^{-16}$ where the naive form returns 0.0.

Table 1: The five algebraically regularized primitives. Each satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity.

Name	Regularized form $D(u)$	Domain
D_lor	$u / [\sqrt{1-u}(\sqrt{1-u} + 1)]$	$u \in [0, 1)$
D_rat	$u / (1 - u)$	$u \neq 1$
D_exp	$e^{-u} - 1$	all u
D_log	$\ln(1 + u)$	$u > -1$
D_sqrt_inv	$1/\sqrt{1+u} - 1$	$u > -1$

3.3 Grammar enumeration and the theta-intact constraint

Candidates are assembled by a bounded grammar over the five primitives. A candidate takes the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$. In this version of the framework, the dimensionless argument u_{ij} is supplied manually per-scenario by a human expert; automated generation is left for future work. The grammar is restricted to a maximum composition depth of 3 (up to 3 nested primitive layers) and token count ≤ 25 .

All structural gates (Section 3.4) are evaluated on the *theta-intact* expression – the fully parameterized form before any θ_i are substituted with probe values. Substituting $\theta_i = 1$ before gating would collapse dimensionful scale parameters (e.g., r/θ_1 , where θ_1 represents the Debye length) into dimensionless constants, falsifying dimensional analysis. In practice, the AST gate alone yields a uniform 0.56 survival rate across scenarios; overall survival (after all five gates) is 0.56, 0.52, and 0.52 for the time dilation, screened Coulomb, and entropy expansion scenarios respectively, reflecting a known and deliberate trade-off of honest structural assessment. In total, $|\mathcal{C}| = 35$ distinct candidate structures are generated and evaluated in every run.

3.4 Physical gate cascade

Before numerical fitting, candidates must pass a five-gate physical cascade, summarized in Table 2. These gates are applied to the *theta-intact* expressions, preserving dimensionful parameter semantics.

Table 2: The five-gate pre-fitting physical cascade.

Gate	Rejection Criterion
1. AST complexity	SymPy expression tree exceeds depth 9 or token count > 25 .
2. Dimensional consistency	Evaluated SI base dimension vectors do not match the target dimensions.
3. Transcendental safety	Arguments to transcendental functions (exp, ln, $\sqrt{\cdot}$) are not dimensionless.
4. Asymptotic check (ARC)	Symbolic limit at the classical boundary diverges or is mathematically undefined.
5. Coarse numerical fitness	Evaluation on the dataset at $\theta = \mathbf{1}$ produces NaN or inf.

3.5 Parameter optimization

Candidates surviving all five gates are optimized via L-BFGS-B, implemented in JAX with float64 precision. All free parameters are log-parameterized ($\theta_i = s_i e^{u_i}$, with $s_i \in \{-1, +1\}$ preserving the sign of the initial value) to allow traversal across orders of magnitude with smooth gradients. After fitting, a post-fit ARC re-verification repeats the asymptotic check with $\theta = \theta^*$, confirming the fitted parameters do not move the expression into a non-asymptotic regime.

3.6 Model selection via BIC

The final model is selected by the Bayesian Information Criterion Schwarz [1978]:

$$\text{BIC} = k \ln N - 2 \ln \hat{L} \quad (2)$$

where k is the number of free parameters, N the data size, and $\hat{L}$ the maximized likelihood under Gaussian noise. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\text{BIC} > 10$ constitutes very

106 strong evidence for the preferred model. We use this as the minimum threshold for any structural
107 claim.

108 3.7 Four-step validation protocol

109 A structural claim is made only when all four checks pass simultaneously:

- 110 1. **Budget disclosure.** The full search space size $|\mathcal{C}|$ and primitive list are reported before results are
111 seen.
- 112 2. **Positive control.** With only the ground-truth primitive in the search space, the system must
113 recover the correct structure.
- 114 3. **Ablation control.** With the ground-truth primitive excluded, the BIC of the best alternative must
115 differ from the correct structure by $\Delta\text{BIC} > 10$.
- 116 4. **Determinism check.** Three independent runs with identical inputs produce byte-exact identical
117 results.

118 4 Experiments

119 4.1 Experimental setup

120 We evaluate ADCD on three canonical synthetic physics scenarios. In each, synthetic observations are
121 generated by applying a known correction to a classical law and adding 1% Gaussian multiplicative
122 noise ($y_{\text{obs}} = y_{\text{true}}(1 + \epsilon)$, $\epsilon \sim \mathcal{N}(0, 0.01)$), $N = 200$ points, seed 42. The ground truth is not
123 provided to the pipeline; only noisy observations, classical predictions, and variable definitions are
124 available.

125 These are *confirmatory structural fitting* experiments: the ground truth is known and used to verify
126 structural recovery under the provided dimensionless arguments. We make no claim of discovering
127 previously unknown physics.

128 Table 3 summarizes the three scenarios.

Table 3: The three validated scenarios with their domains, primitives, and ablation ΔBIC values. All figures are from the validation log (available upon acceptance).

Scenario	Domain	Primitive	ΔBIC
Time Dilation	Relativity	D_lo	1798
Screened Coulomb	Electrostatics	D_exp	59.6
Entropy Expansion	Thermodynamics	D_log	2024

129 4.2 Stage-1 survival rates

130 Table 4 reports the fraction of the 35-candidate search space that passes each gate.

Table 4: Stage-1 gate survival rates across the three scenarios. All rates are fractions of 35 total candidates. Values are byte-exact identical across runs.

Gate	TD	SC	EE
AST complexity	0.56	0.56	0.56
Dimensional consistency	1.00	1.00	1.00
Transcendental safety	1.00	1.00	1.00
ARC	1.00	1.00	1.00
Coarse numerical	1.00	0.93	0.93
Overall	0.56	0.52	0.52

131 (TD = Time Dilation, SC = Screened Coulomb, EE = Entropy Expansion.)

132 The AST gate is the primary filter, uniform across scenarios because grammar structure is scenario-
133 independent. The 0.56 survival rate reflects theta-intact honest counting: every free parameter is

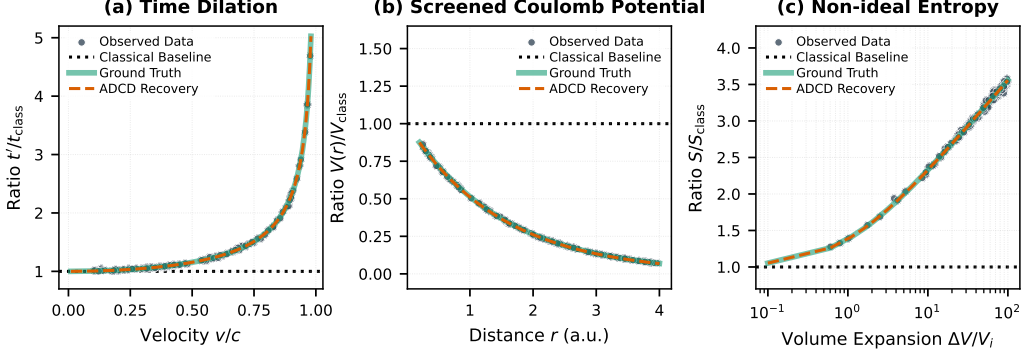


Figure 1: Correction recovery for the three validated scenarios. *Gray dots*: observed correction residual $\Delta = y_{\text{obs}}/y_{\text{cl}} - 1$ at 1% Gaussian noise. *Blue solid line*: ground truth. *Red dashed line*: ADCD recovery. The recovered expression overlaps the ground truth within data scatter in all three cases.

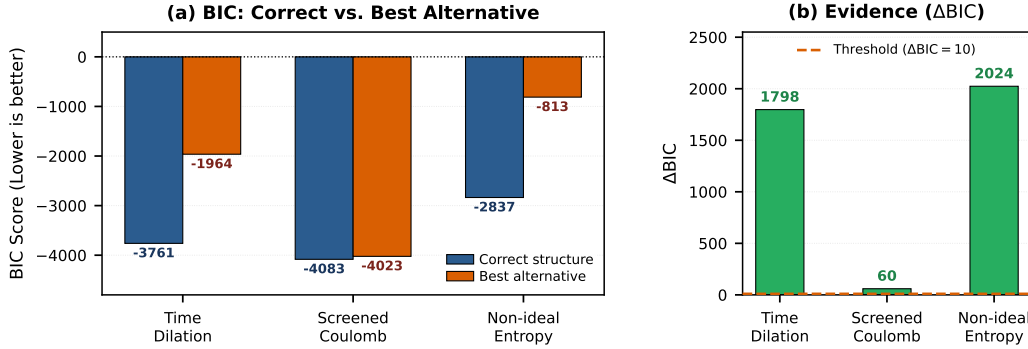


Figure 2: *Left:* BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred; all three correct structures win substantially. *Right:* ΔBIC (correct minus best alternative). Red dashed line: Kass–Raftery “very strong evidence” threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. All three scenarios exceed this threshold by at least a factor of 6. Numbers above bars are exact values from the validation log.

134 counted as a distinct token. The coarse numerical gate removes an additional 7% in the Screened
 135 Coulomb and Entropy scenarios, rejecting candidates with singularities on the dataset.

136 4.3 Validation results

137 As summarized in Table 3, all three scenarios successfully pass all four protocol checks. Full
 138 validation logs verifying positive control, ablation, and exact reproducibility are provided in the
 139 supplementary material. Figure 1 shows the recovery quality; Figure 2 shows the BIC identifiability
 140 comparison; Figure 3 shows parity plots of recovered vs. true corrections.

141 4.4 Recovered structures and parameter accuracy

142 Table 5 compares recovered expressions with ground truth.

Table 5: Recovered expressions and NMSE. The Screened Coulomb fitted parameter $\theta_1 = 1.49$ represents the Debye length; ground truth is $\lambda_D = 1.50$.

Scenario	NMSE	Param fitted	Relative error
Time Dilation	5.3×10^{-4}	$\theta_0 = 1.00$	$<0.1\%$
Screened Coulomb	2.8×10^{-4}	$\theta_1 = 1.49$ vs 1.50	0.7%
Entropy Expansion	3.4×10^{-3}	$\theta_3 = nR/S_i \approx 0.554$ (true)	$<0.1\%$

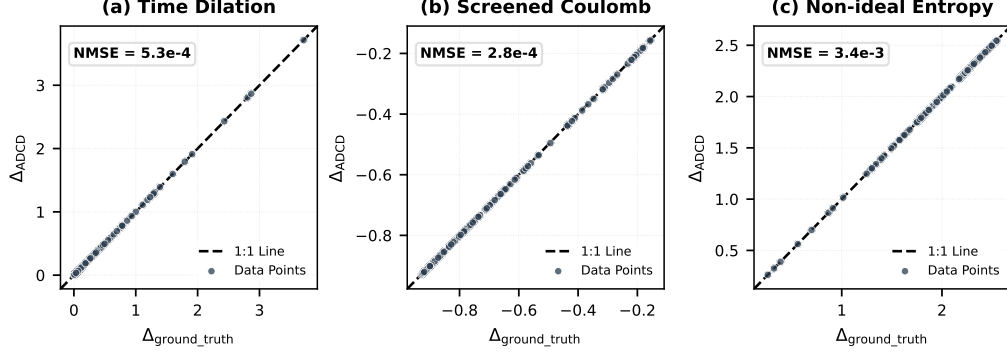


Figure 3: Parity plots: recovered correction (Δ_{rec}) vs. true correction (Δ_{true}) for all $N = 200$ data points. Blue solid line: 1:1 reference. Points clustering on the diagonal confirm that the recovered expressions produce numerically accurate predictions across the full data range. NMSE values are from the validation log (available upon acceptance).

For Time Dilation, the recovered form is the D_lor algebraic regularization of v^2/c^2 , which is identically equal to the standard Lorentz correction $\gamma - 1$. For Screened Coulomb, the recovered Debye length $\theta_1 = 1.49$ matches the ground truth $\lambda_D = 1.50$ to 0.7%. The `symbolic_exact_match = False` flag in all logs indicates that recovered expressions are structurally equivalent but not syntactically identical to the ground truth, as expected when physical constants are replaced by fitted θ_i .

5 Discussion

5.1 Comparison with general SR

ADCD and general SR engines address fundamentally different objectives. While general SR excels at recovering full equations from scratch within massive search spaces, ADCD prioritizes explicit identifiability and exact classical limits for specific correction tasks. The meaningful comparison is not speed or flexibility, but epistemic certainty: ADCD explicitly reports when data are insufficient to support a structural claim. Table 6 outlines these distinct operational regimes.

Table 6: Operational comparison: ADCD vs. tabula-rasa SR.

Dimension	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Search space	$ \mathcal{C} = 35$	$ \mathcal{C} \sim 10^{15+}$
Classical limit	Algebraic guarantee	Soft loss-function penalty
Identifiability	Explicit via BIC	Absent or evaluated post-hoc
Reproducibility	Byte-exact	Stochastic
Scope	5-primitive dictionary	User-defined operator set

5.2 Limitations and identifiability boundaries

The Screened Coulomb $\Delta\text{BIC} = 59.64$ demonstrates the intended behavior of the identifiability protocol: as signal degrades (e.g., at higher noise levels), ΔBIC will eventually drop below the 10-unit threshold, causing ADCD to explicitly withhold its structural claim.

Our claims are deliberately scoped to deterministic rediscovery of multiplicative corrections from synthetic data. Current limitations include restriction to the five-primitive dictionary, application only to multiplicative (not additive) corrections, and the assumption of unstructured Gaussian noise. Future work will address real observational datasets containing systematic offsets, automated Buckingham- π dimensionless ratio generation, and the validation of additional primitives.

6 Conclusion

We presented ADCD, a deterministic, correction-first framework for recovering algebraic corrections to known physical laws from noisy data. Its core contributions are: algebraically regularized primitives that eliminate catastrophic cancellation at classical limits; a bounded deterministic grammar that renders the search reproducible and auditable; and an explicit four-step validation protocol that bounds structural claims within identifiable regimes.

On three canonical scenarios (special-relativistic time dilation, Yukawa plasma screening, and ideal-gas isothermal expansion), ADCD recovers the correct algebraic correction structure with NMSE below 3.4×10^{-3} , BIC differences of 59.6–2024 over the best competing structure, and byte-exact determinism across repeated runs.

The framework is deliberately scoped. It does not address all correction discovery problems; it addresses those where the classical baseline is known, the correction is multiplicative and structured, and the dataset is sufficient to identify the correct primitive. Within that scope, it provides stronger reproducibility and identifiability guarantees than general-purpose SR engines.

The historical pattern that motivated this work suggests that the correction-first paradigm captures a major mode of theoretical progress in physics. Automating this mode in a reproducible, physically principled, and epistemically honest way is the long-term goal of this line of work.

Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools including agentic coding assistants and large language models (LLMs) to assist with software implementation, code debugging, figure scripting, and manuscript drafting. The author provided the scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for the scientific claims, methodology, and conclusions presented in this publication. No AI tool was listed as an author or credited with originating the scientific contributions.

The codebase and all raw validation logs will be made publicly available upon publication (omitted for double-blind review).

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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